

Booting Process , Port Numbers ,Linux Server Administration

Questions :

- Booting Process in Linux (Steps in detail)
- Important Port Numbers
- Some Advanced Server Administrator commands for troubleshooting and monitoring

1. Booting Process in Linux

The Linux boot process is the sequence of operations that takes place from powering on the system until the operating system becomes ready for use.

Boot Process Flow

```
Power ON
↓
BIOS / UEFI
↓
GRUB Bootloader
↓
Linux Kernel
↓
initramfs
↓
systemd (PID 1)
↓
systemd Targets
↓
System Services
↓
Login / System Ready
```

Step 1: BIOS / UEFI

When the computer is powered on, **BIOS or UEFI firmware** starts first.

Main functions:

- Performs hardware initialization.
- Checks and initializes RAM, CPU, storage, and other devices.
- Finds a bootable device.
- Loads the bootloader.

Modern systems commonly use **UEFI**.

Step 2: GRUB Bootloader

After BIOS/UEFI, the **GRUB 2 bootloader** is loaded.

GRUB is responsible for:

- Displaying available operating systems or kernels.
- Selecting the Linux kernel to boot.
- Loading the Linux kernel into memory.
- Passing kernel parameters.
- Loading the initial RAM filesystem.

Step 3: Linux Kernel

GRUB loads the Linux kernel into memory.

The kernel is the core component of Linux and is responsible for managing:

- CPU
- Memory
- Processes
- Hardware devices
- Networking
- Filesystems

Step 4: initramfs

initramfs stands for **initial RAM filesystem**.

It provides the temporary environment and tools required during the early boot process.

It may contain:

- Storage drivers
- Filesystem drivers
- LVM support
- RAID support
- Encryption support
- Tools required to locate and mount the root filesystem

Step 5: systemd

After the kernel has initialized the system, it starts **systemd**.

On modern Rocky Linux/RHEL systems, systemd is normally the first userspace process and has:

PID = 1

systemd is responsible for:

- Starting services
- Managing services
- Managing targets
- Handling system startup
- Managing system shutdown
- Managing systemd units

Step 6: systemd Targets

systemd starts the appropriate **target** for the system.

Common targets include:

- `multi-user.target`
- `graphical.target`

`multi-user.target` is commonly used for server environments.

`graphical.target` is used when a graphical desktop environment is required.

Step 7: System Services

systemd starts the services required by the system and those configured to start automatically.

Common services include:

- SSH
- NetworkManager
- firewalld
- chronyd
- Apache
- Nginx

Step 8: Login / System Ready

After the required system components and services have started, the system becomes ready.

Users can access the system through:

- Console
- SSH
- Graphical login

2. Important Port Numbers

A **port number** identifies a specific network service or application endpoint on a computer.

Port	Protocol / Service	Purpose
20	FTP	FTP data transfer
21	FTP	FTP control
22	SSH	Secure remote administration
23	Telnet	Remote login; insecure
25	SMTP	Sending/relaying email
53	DNS	Domain name resolution
67	DHCP	DHCP server

68	DHCP	DHCP client
80	HTTP	Web traffic
110	POP3	Email retrieval
123	NTP	Time synchronization
143	IMAP	Email retrieval
161	SNMP	Network monitoring
389	LDAP	Directory services
443	HTTPS	Secure web traffic
445	SMB	File sharing
636	LDAPS	Secure LDAP
3306	MySQL/MariaDB	Database
5432	PostgreSQL	Database
6379	Redis	Database/cache
8080	HTTP alternative	Common application/web port
8443	HTTPS alternative	Common application port

3. Advanced Server Administrator Commands for Troubleshooting and Monitoring

3.1 systemctl

systemctl is used to manage **systemd services and units**.

Common operations:

```
systemctl status <service>
sudo systemctl start <service>
sudo systemctl stop <service>
sudo systemctl restart <service>
sudo systemctl enable <service>
sudo systemctl enable --now <service>
systemctl is-enabled <service>
```

3.2 journalctl

journalctl is used to view logs collected by the **systemd journal**.

Common operations:

```
journalctl -n 50
journalctl -f
journalctl -u <service>
journalctl -u <service> -n 30 --no-pager
journalctl -b
journalctl -b -1
```

3.3 ss

ss is used to display network sockets and listening ports.

```
ss -tulnp
```

Options:

```
-t → TCP
-u → UDP
-l → Listening
-n → Numeric output
-p → Process information
```

3.4 ip a

ip a is used to display network interfaces and IP addresses.

```
ip a
```

It provides information about:

- Network interfaces
- IP addresses
- Subnet prefixes
- Interface states

3.5 ip route

`ip route` displays the system's routing table.

`ip route`

It provides information about:

- Default gateway
- Network routes
- Network interfaces associated with routes

3.6 ping

`ping` is used to test basic network connectivity.

`ping <IP-address>`

It is useful for checking whether a destination is reachable over the network.

3.7 curl

`curl` is a command-line tool used to communicate with network services, particularly HTTP/HTTPS services.

`curl localhost`

`curl -I http://localhost`

It is useful for testing web servers and checking HTTP responses.

3.8 ps

`ps` is used to display running processes.

`ps aux`

`ps -ef`

It provides information about:

- Process IDs
- Users
- CPU usage
- Memory usage
- Running processes

3.9 top

top provides real-time information about processes and system resource usage.

top

It can monitor:

- CPU usage
- Memory usage
- Processes
- Load
- Process IDs

3.10 htop

htop is an interactive process viewer that provides a more user-friendly view of system processes and resource usage.

htop

It can monitor:

- CPU usage
- Memory usage
- Processes
- Process IDs
- Resource consumption

3.11 free

free displays memory and swap usage.

`free -h`

It is useful for monitoring:

- Total memory
- Used memory
- Available memory
- Free memory
- Swap usage

3.12 df

`df` displays filesystem disk-space usage.

`df -h`

It is useful for monitoring:

- Filesystem size
- Used space
- Available space
- Disk utilization percentage

3.13 du

`du` is used to determine the disk space consumed by files and directories.

`du -sh <directory>`

`du -sh /var/*`

It is useful for identifying directories that are consuming large amounts of disk space.

3.14 uptime

`uptime` displays information about system uptime and load.

`uptime`

It provides:

- Current system uptime
- Number of logged-in users
- Load average

3.15 vmstat

vmstat provides information about system performance.

```
vmstat
```

```
vmstat 2
```

It provides information related to:

- Processes
- Memory
- Swap
- I/O
- CPU

3.16 iostat

iostat is used to monitor CPU and disk I/O performance.

```
iostat
```

```
iostat -xz 2
```

It is useful for identifying storage and I/O-related performance issues.

3.17 dig

dig is a DNS troubleshooting tool.

```
dig <domain>
```

```
dig <domain> A
```

It is used to examine DNS resolution and DNS records.

3.18 lsof

`lsof` means **List Open Files**.

It can be used to identify processes using files, devices, and network ports.

```
sudo lsof -i :<port>
```

It is particularly useful for identifying which process is using a specific network port.

3.19 dmesg

`dmesg` displays messages from the Linux kernel.

```
dmesg
```

```
dmesg | tail -30
```

It can be useful for investigating:

- Hardware problems
- Drivers
- Storage devices
- Network interfaces
- Kernel-related issues